FREEDOM INTERNATIONAL SCHOOL

WORKSHEET

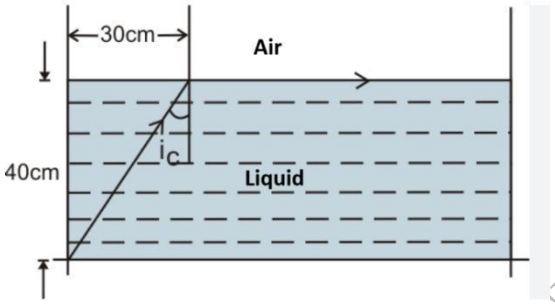
PHYSICS

CLASS XII

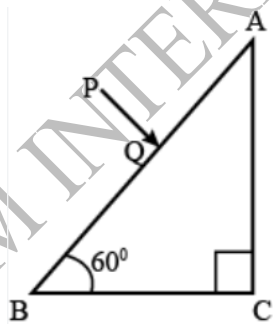
RAY OPTICS AND OPTICAL INSTRUMENTS

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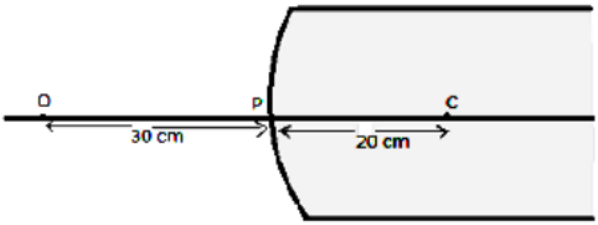
1. Calculate the speed of light in a medium, whose critical angle is 30^0.
2. In the following ray diagram, calculate the speed of light in the liquid of unknown refractive index.



3. A ray PQ incident normally on the refracting face BA is refracted in the prism BAC made of material of refractive index 1.5. Complete the path of ray through the prism. From which face will the ray emerge? Justify your answer.

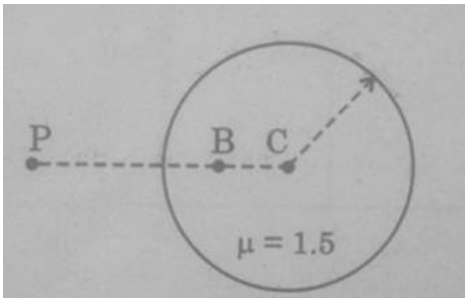


4. A spherical convex surface of radius of curvature 20 cm, made of glass (μ= 1.5) is placed in air. Find the position of the image formed, if a point object is placed at 30 cm in front of the convex surface on the principal axis.

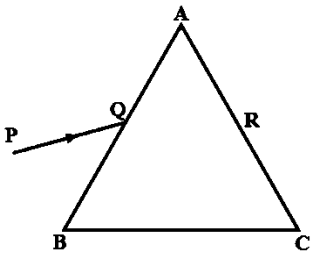


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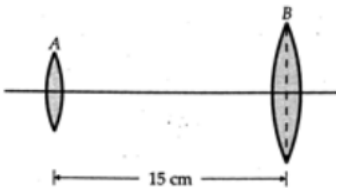
5. An air bubble is trapped at point B (CB= 20 cm) in a glass sphere of radius 40 cm and refractive index 1.5 as shown in the figure. Find the nature and position of the image of the bubble as seen by an observer at point P.



6. A convex lens of focal length 0.2 m and made of glass (μ= 1.50) is immersed in water (μ= 1.33). Find the change in the focal length of lens.
7. A plano- convex lens μ= 1.5 has focal length of 18 cm in air. Calculate the radius of curvature of spherical surface.
8. A convex lens of focal length 10 cm is placed coaxially 5 cm away from a concave lens of focal length 10 cm. If an object is placed 30 cm in front of the convex lens, find the position of the final image formed by the combined system.
9. A ray of light incident on a face of an equilateral prism shows minimum deviation of 30^0. Calculate the speed of light through the prism.
10. A ray PQ incident on the face AB of a prism ABC, as shown in the figure, emerges from the face AC such that AQ= AR. Draw the ray diagram showing the passage of the ray through the prism. If the angle of the prism is 60^0 and refractive index of the material of the prism $\sqrt{3}$, determine the values of the angle of incidence and angle of deviation.



11. The total magnification produced by a compound microscope is 20, while that produced by the eyepiece alone is 5. When the microscope is focused on a certain object, the distance between objective and eyepiece is 14 cm. Find the focal length of objective and eyepiece, if the distance of distinct vision is 20 cm.
12. In normal adjustment, for a refracting telescope, the distance between the objective and eyepiece lens is 1.00 m. If the magnifying power of the telescope is 19, find the focal length of the objective and the eyepiece.
13. Two convex lenses A and B of an astronomical telescope having focal lengths 5 cm and 20 cm respectively, are arranged as shown in the figure.
    (i) Which one of the two lenses will you select to use as the objective lens and why?
    (ii) What should the change in the distance between the lenses be to have the telescope in its normal adjustment position?
    (iii) Calculate the magnifying power of the telescope in the normal adjustment position.



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